

BRIGHTLEARN DATA ANALYTICS

SQL FUNDAMENTALS EXERCISE 3

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Course: BrightLearn Basic SQL

SQL CASE Statements

Section 1 Q1. Classify each product by price into the three tiers below.

Product_id	Product_name	Price	'Expensive' if price > 1000 'Mid-range' if price between 100 and 1000 'Budget' if price < 100
1	Laptop	1200.00	
2	Phone	800.00	
3	Keyboard	45.00	
4	Monitor	300.00	
5	Mouse	25.00	

```
SELECT product_name,  
       price,
```

```
CASE
```

```
  WHEN price > 1000 THEN 'Expensive'
```

```
  WHEN price BETWEEN 100 AND 1000 THEN 'Mid-range'
```

```
  WHEN price < 100 THEN 'Budget'
```

```
END AS price_category
```

```
FROM products;
```

Products

Product_name	Price	price_category
Laptop	1200	Expensive
Phone	800	Mid-range
Keyboard	45.00	Budget
Monitor	300.00	Mid-range
Mouse	25.00	Budget

Q2 Label each order by its value.
orders

Customer_name	amount	order_value_category
Alice	150.00	Low Value
Bob	560.00	Medium Value
Charlie	999.99	Medium Value
Diana	45.50	Low Value
Ethan	1200.00	High Value

```
SELECT customer_name,  
       amount,
```

```
CASE
```

```
  WHEN amount >= 1000 THEN 'High Value'
```

```
  WHEN amount BETWEEN 500 AND 999.99 THEN 'Medium  
Value'
```

```
  WHEN amount < 500 THEN 'Low Value'
```

```
END AS order_value_category
```

```
FROM orders;
```

Q3. Categorise

Q3. Categorise each employee's position level using both department and salary.

- If in 'IT' and salary > 80000 → 'Senior IT'
- If in 'HR' and salary > 55000 → 'Experienced HR'
- Otherwise - 'Staff'

CASE STATEMENT on the next page →→→

```
SELECT emp_name, department, salary,
CASE
```

```
  WHEN department = 'IT' AND salary > 80000 THEN 'Senior IT'
```

```
  WHEN department = 'HR' AND salary > 55000 THEN 'Experienced HR'
```

```
  ELSE 'staff'
```

```
END AS position_level
```

```
FROM employees;
```

employees

emp_name	department	salary	Position_level
John	IT	85000	Senior IT
Sara	HR	60000	Experienced HR
Mark	IT	75000	Staff
Lucy	Finance	95000	Staff
Tom	HR	55000	Staff

Q4 Assign each student a letter grade based on their
Score

```
SELECT student_name, score,
```

```
CASE
```

```
  WHEN score ≥ 90 THEN 'A'
```

```
  WHEN score BETWEEN 80 AND 89 THEN 'B'
```

```
  WHEN score BETWEEN 70 AND 79 THEN 'C'
```

```
  WHEN score BETWEEN 60 AND 69 THEN 'D'
```

```
  WHEN score < 60 THEN 'F'
```

```
END AS grade
```

```
FROM students;
```

Students

student_name	Score	grade
Anna	92	A
Ben	76	C
Cara	59	F
David	83	B
Ella	68	D

$\geq 90 \rightarrow 'A'$
 $80-89 \rightarrow 'B'$
 $70-79 \rightarrow 'C'$
 $60-69 \rightarrow 'D'$
 $< 60 \rightarrow 'F'$

Q5 Label delivery performance based on the time taken.

SELECT delivery_id, delivery_time_minutes,

CASE

WHEN delivery_time_minutes ≤ 30 THEN 'Fast'

WHEN delivery_time_minutes BETWEEN 31 AND 60 THEN 'On Time'

WHEN delivery_time_minutes > 60 THEN 'Late'

END AS performance

FROM deliveries;

deliveries

delivery_id	delivery_time_minutes	performance
1	45	ON Time
2	80	Late
3	30	Fast
4	65	Late
5	100	Late

$\leq 30 \text{ mins} \rightarrow 'Fast'$
 $31-60 \text{ mins} \rightarrow 'ON Time'$
 $> 60 \text{ mins} \rightarrow 'Late'$

Q6. Convert the numeric priority into a readable label.

3 $\rightarrow$ 'High'

2 $\rightarrow$ 'Medium'

1 $\rightarrow$ 'Low'

Tickets

issue_type	priority	priority_label
Login issue	1	Low
Server down	3	High
Slow system	2	Medium
Email error	2	Medium
Password reset	1	Low

```
SELECT issue_type, priority;  
CASE  
    WHEN priority = 3 THEN 'High'  
    WHEN priority = 2 THEN 'Medium'  
    ELSE 'Low'  
END AS priority_label  
FROM tickets;
```

Q7. Calculate attendance percentage and classify the result.

```
SELECT student_id,  
    days_present / total_days * 100 AS attendance_percentage,  
CASE  
    WHEN days_present / total_days * 100 ≥ 90 THEN 'Excellent'  
    WHEN days_present / total_days * 100 BETWEEN 75 AND 89 THEN  
        'Good'  
    WHEN days_present / total_days * 100 < 75 THEN 'Needs Improvement'  
END AS attendance_status  
FROM attendance;
```

- * $\geq 90\%$ → 'Excellent'
- * $75-89\%$ → 'Good'
- * $< 75\%$ → 'Needs Improvement'

attendance

Student_id	attendance_percentage	attendance_status
1	90	Excellent
2	60	Needs Improvement
3	96	Excellent
4	50	Needs Improvement
5	100	Excellent

Q8. Label the stock status of each product

```
SELECT product_id, stock_qty,
CASE
```

```
WHEN stock_qty = 0 THEN 'Out of Stock'
```

```
WHEN stock_qty BETWEEN 1 AND 5 THEN 'Low Stock'
```

```
WHEN stock_qty > 5 THEN 'In Stock'
```

```
END AS stock_status
```

```
FROM inventory;
```

Products_inventory

product_id	stock_qty	stock_status
1	5	Low Stock
2	0	Out of Stock
3	25	In Stock
4	10	In Stock
5	3	Low Stock

Q9. Classify each class by the number of enrolled students

$\geq 25 \rightarrow$ 'Large'

$10 - 24 \rightarrow$ 'Medium'

$< 10 \rightarrow$ 'Small'

```

SELECT subject, enrolled_students,
CASE
    WHEN enrolled_students ≥ 25 THEN 'Large'
    WHEN enrolled_students BETWEEN 10 AND 24 THEN 'Medium'
    WHEN enrolled_students < 10 THEN 'Small'
END AS class_size_category
FROM classes;

```

Classes

Subject	enrolled_students	class_size_category
Math	30	Large
English	25	Large
Science	15	Medium
Art	5	Small
History	20	Medium

Q10. Apply a discount flag based on the payment method & amount.

```

SELECT payment_id, payment_method, amount,
CASE
    WHEN payment_method = 'Cash' AND amount ≥ 200 THEN
        'Eligible for Discount'
    ELSE 'Not Eligible'
END AS discount_eligibility

```

FROM payments;

Payments

payment_id	payment_method	amount	discount_eligibility
1	Card	50.00	Not Eligible
2	Cash	200.00	Eligible for Discount
3	Card	150.00	Not Eligible
4	PayPal	75.00	Not Eligible
5	Cash	300.00	Eligible for Discount